

E9: 309 Advanced Deep Learning

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Schedule - MW - 430-6pm (Microsoft Teams)
<http://leap.ee.iisc.ac.in/sriram/teaching/ADL2020/>

The bell has rung



<http://phdcomics.com/>

Recap of deep learning



Housekeeping

✳ Filling the google form in the webpage

➡ Contents will be made available to the folks in the mailing lists.

✓ Announcements regarding evaluations and projects will be shared only with creditors as well as video links.

★ Teams channel interaction and TA session for creditors only.

✳ Online registration portal from academics.iisc.ac.in

✓ Your research/faculty advisor may need to approve also before the deadline (Oct. 20th?)



Some notations

✱ $x \in \mathcal{R}^D$ - input data.

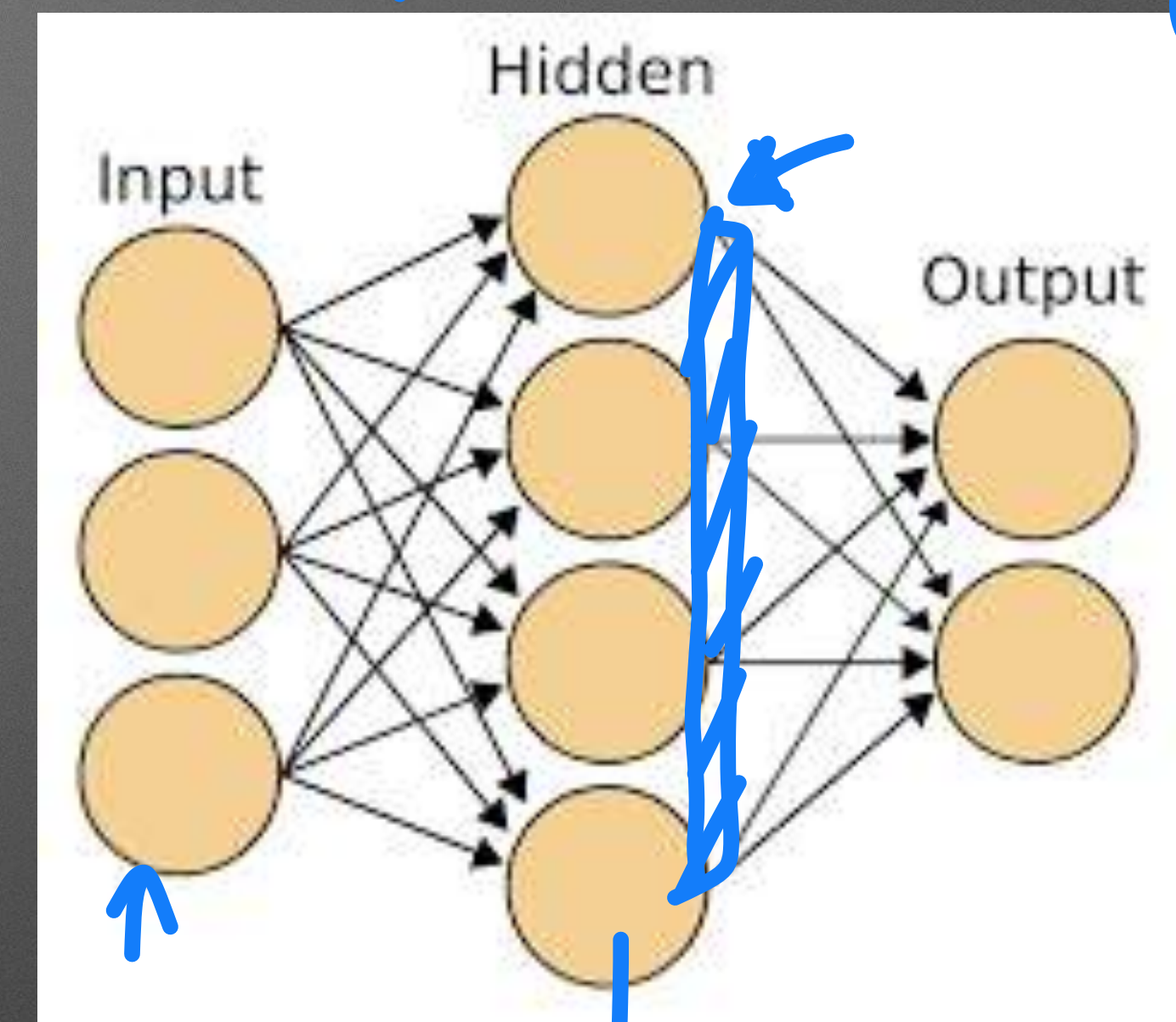
✱ $y \in \mathcal{R}^C$ - neural network targets.

✱ $\hat{y} \in \mathcal{B}^C$ - model outputs.

✱ $e, h \in \mathcal{R}^d$ - hidden model representations or embeddings.

✱ Θ - collection of learnable parameters in the model.

✱ $E(y, \hat{y})$ - error function used in the model training.



Some notations

✳ $\{\mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{y}_1, \dots, \mathbf{y}_N\}$ - labeled training data

✳ $q = \{1 \dots Q\}$ - iteration index.

✳ $t = \{1 \dots T\}$ - discrete time index.

✳ $l = \{1 \dots L\}$ - layer index

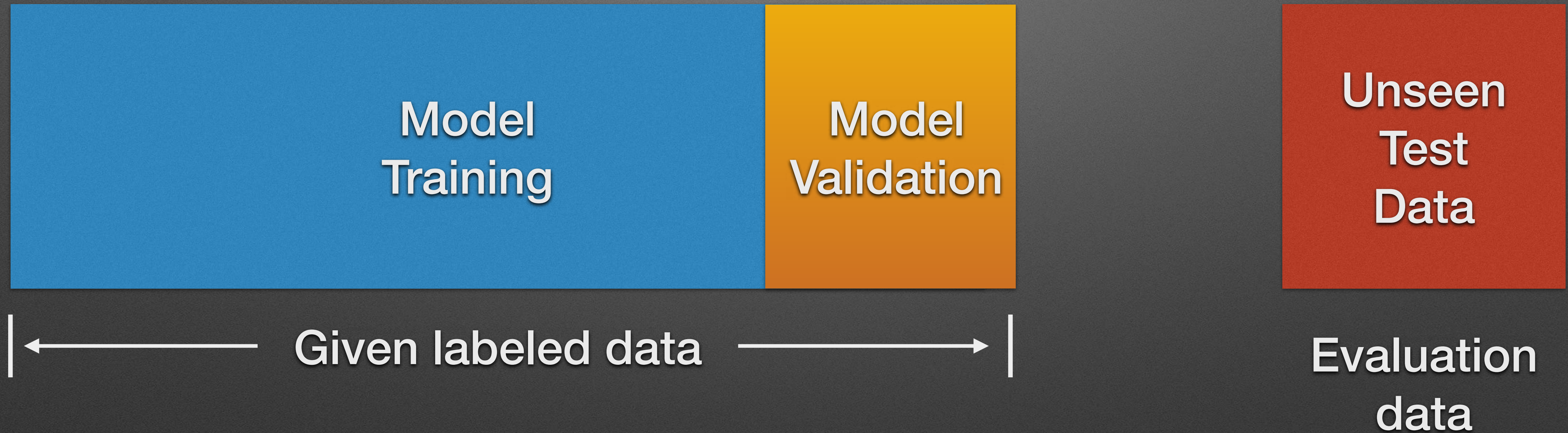
✳ η - learning rate (hyper-parameter)

✳ N_b - mini-batch size and B is the number of mini-batches.



Premise

✳ Training data, validation data, test data.



✳ Model training data - used for parameter learning.

✳ Validation data - used for hyper-parameter tuning (cross validation CV).



Last lecture

- ✳ Feedforward networks
- ✳ Convolutional networks
- ✳ Learning in deep networks
 - ➔ Parameters, hyper-parameters etc
 - ➔ SGD learning rule
 - ➔ Adam optimization, batch normalization.
- ✳ Reading assignments.



Module - I Visual and Time Series Modeling



Why do need recurrent models

✱ Learn from ordered pairs of x, y

✓ All the data samples are treated independently.

★ Data are shuffled before mini-batch formation

✱ If the input data and output labels are time-series data $x(t), y(t)$

◉ DNNs/CNNs may fail to model the correlation of the data across the time

◉ **Question** - how can we build models that capture the time evolution of the data and the labels.



Why do need recurrent models

✳ An interesting subset of this problem is where the input alone is a time series $x(t)$, y or have different indices $x(t)$, $y(u)$

✳ Examples

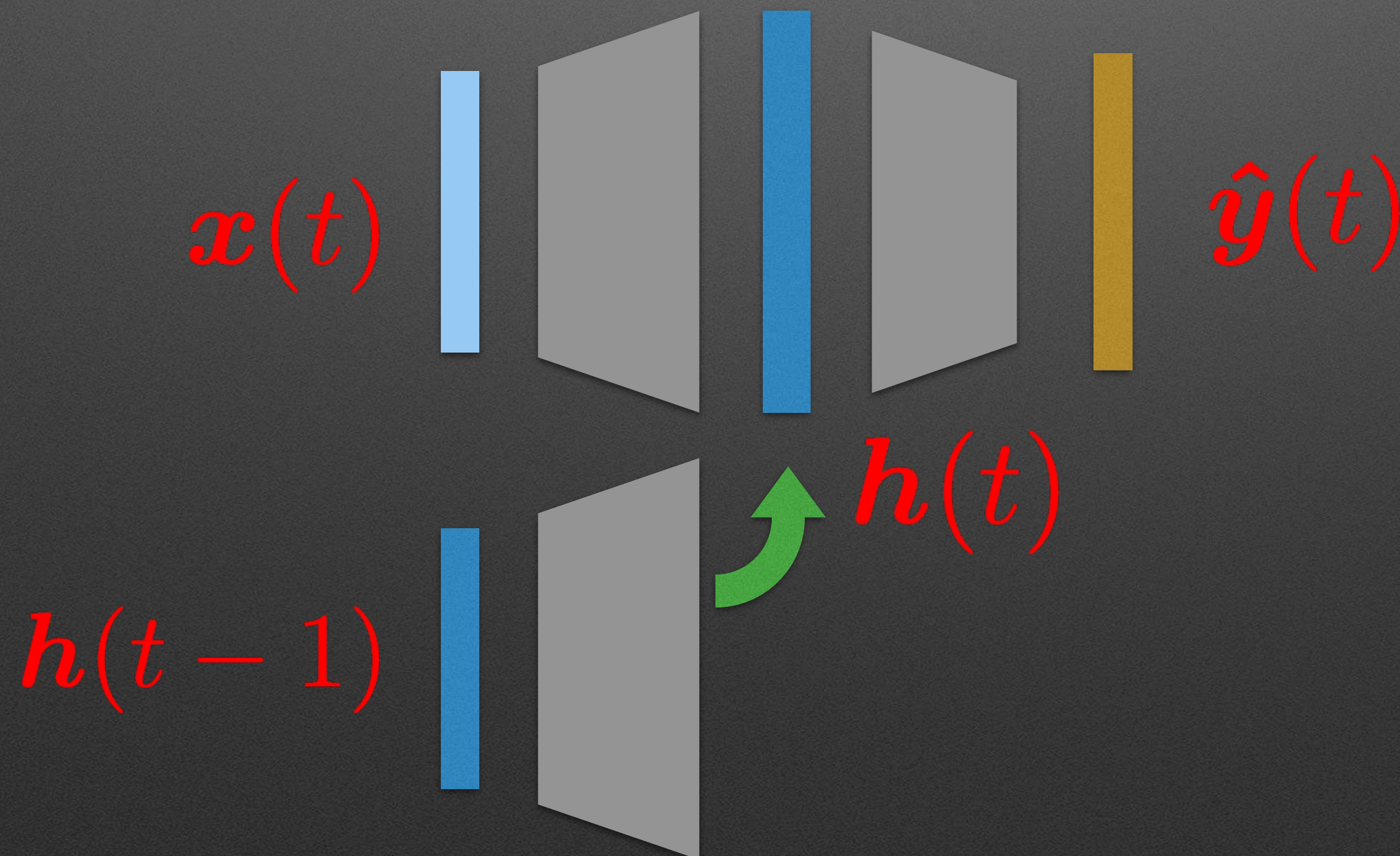
- ✓ Text sequences
- ✓ Speech and audio
- ✓ Video sequences



First order recurrence - hidden layer

- ✳ Making the hidden layer a function of the previous outputs from the hidden layer along with the input

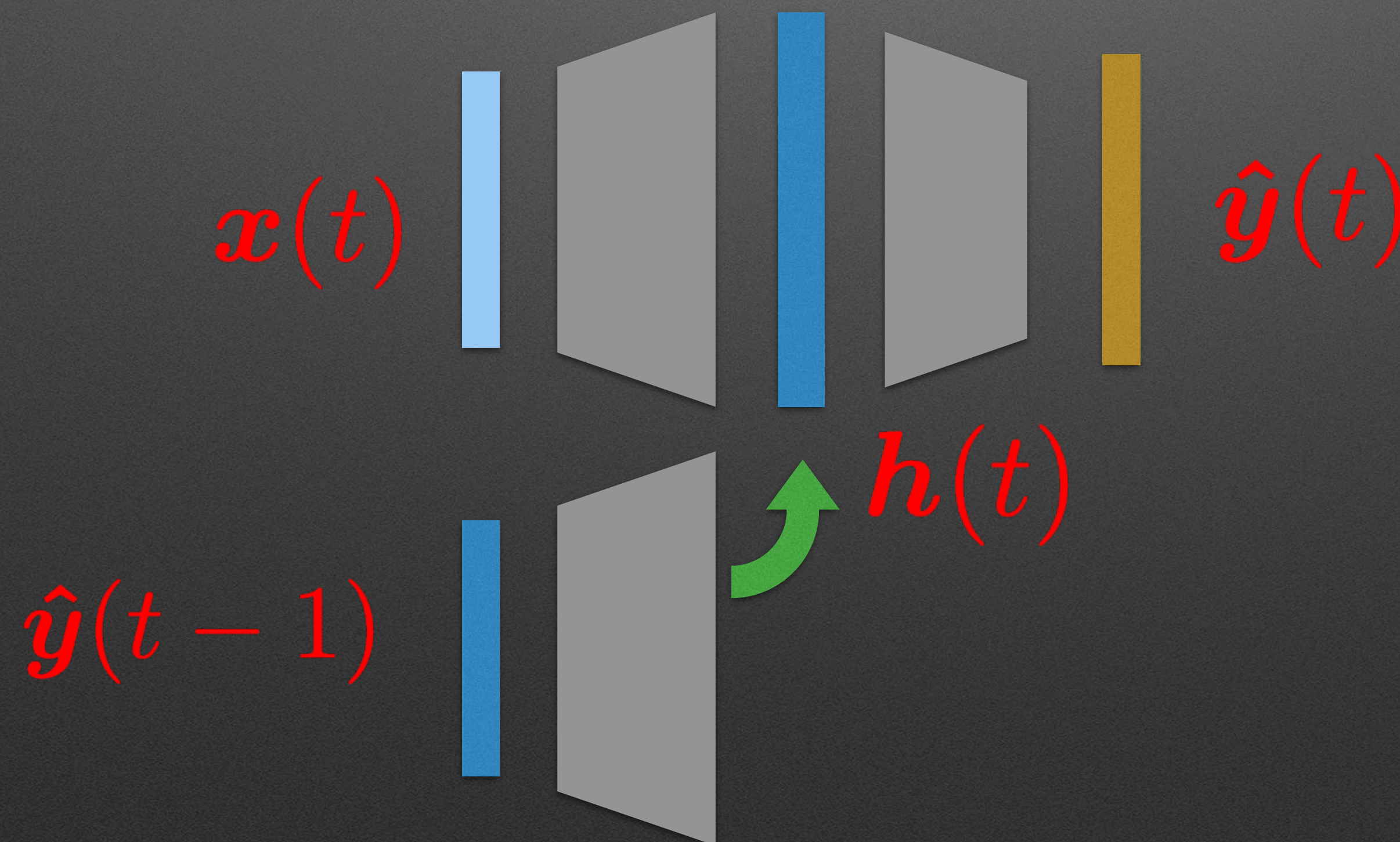
$$h(t) = f(h(t-1), x(t))$$



First order recurrence - output layer

- ✳ Making the hidden layer a function of the previous outputs from the output layer along with the input

$$h(t) = f(\hat{y}(t-1), x(t))$$

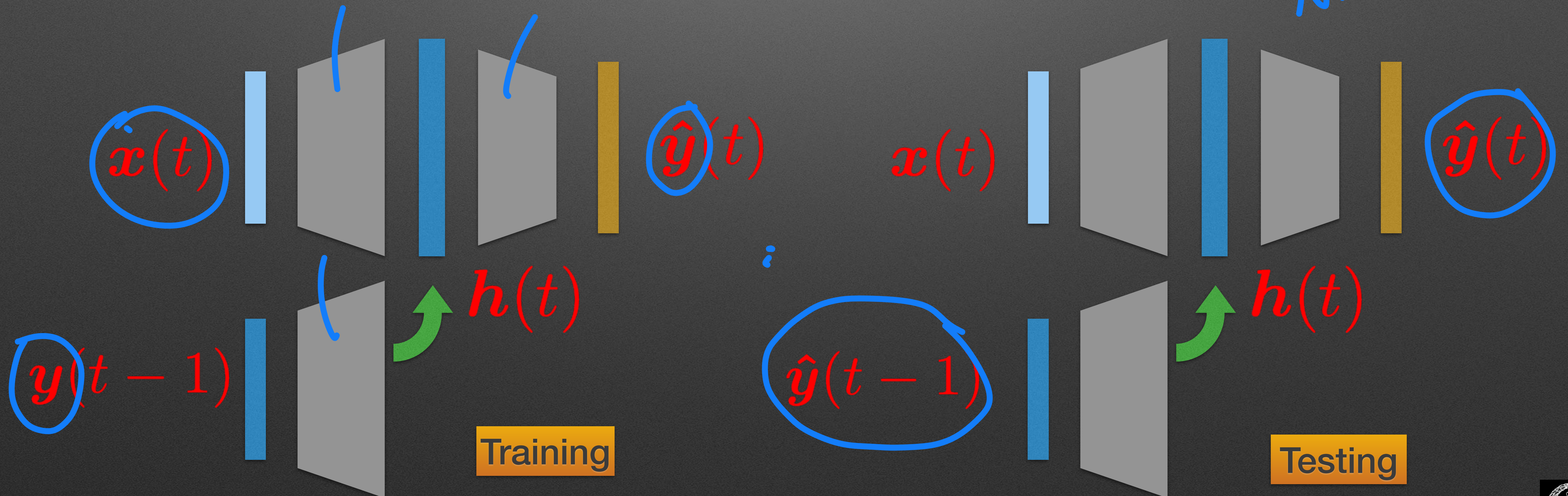


First order recurrence - output layer

- * Making the hidden layer a function of the previous outputs from the output layer along with the input

$$h(t) = f(\hat{y}(t-1), x(t))$$

Teacher Forcing
Network



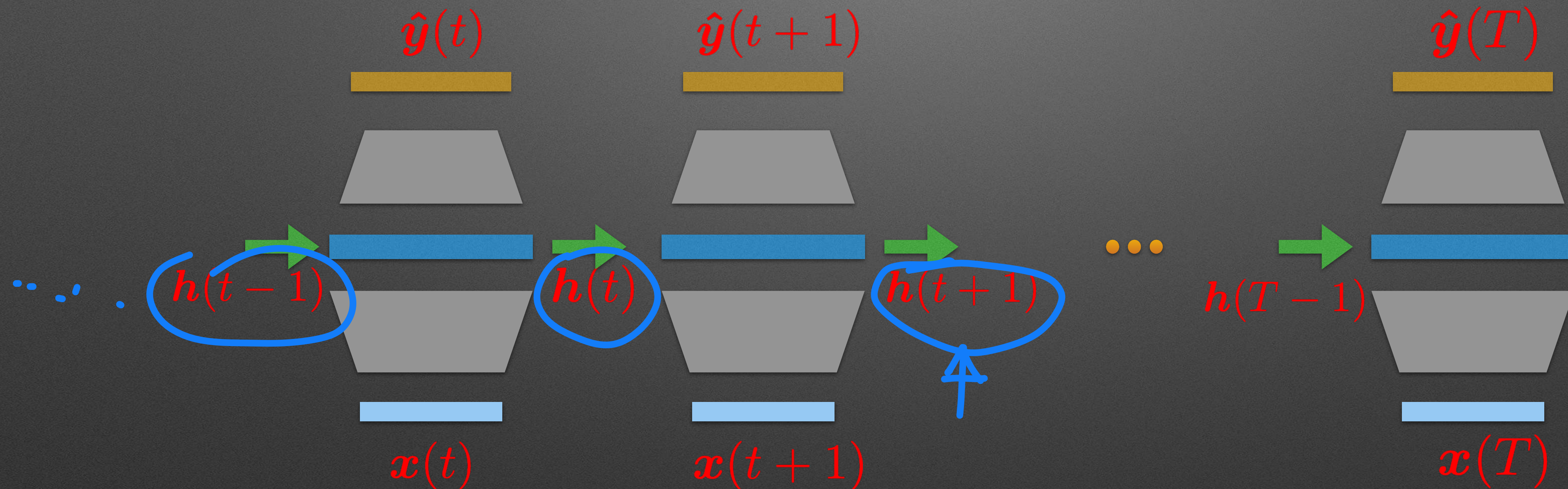
Coming up ...

- ✳ Learning in recurrence networks: Back-propagation in time.
- ✳ Learning considerations : Issues with forgetting and long-short-term memory networks



First order recurrence - hidden layer

- ✳ Making the hidden layer a function of the previous outputs from the hidden layer along with the input.



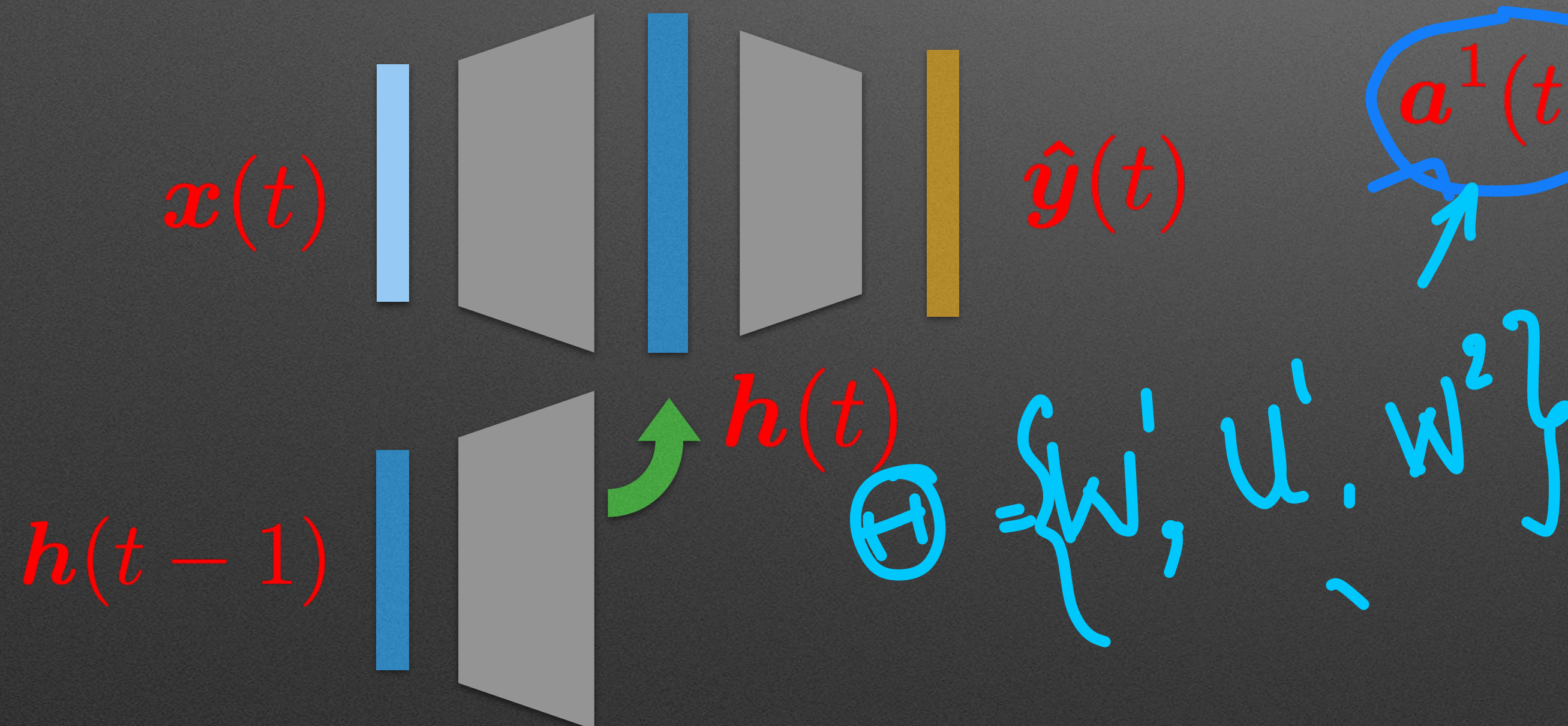
- ✳ Makes the hidden layer dependent of previous layer outputs in a recurring fashion.

First order recurrence - hidden layer

- ✳ Making the hidden layer a function of the previous outputs from the hidden layer along with the input.

$$\underline{h(t)} = \underline{f(h(t-1), x(t))}$$

Model Forward Pass - 1 hidden layer

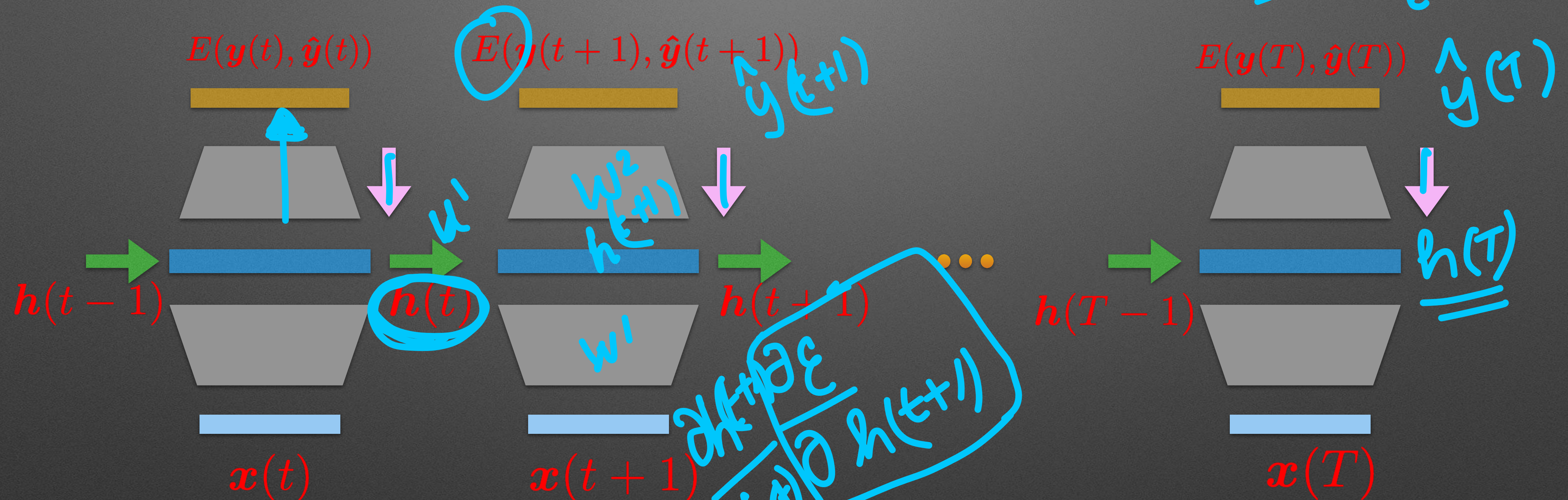


$$\begin{aligned} a^1(t) &= \underline{W^1} x(t) + \underline{U^1} h^1(t-1) \\ h^1(t) &= \tanh(a^1(t)) \\ a^2(t) &= \underline{W^2} h^1(t) \\ \hat{y}(t) &= \underline{S}(a^2(t)) \\ &\quad \text{Softmax} \end{aligned}$$

- ✳ Makes the hidden layer dependent of previous layer outputs in a recurring fashion.

Error backpropagation

✳ Error functions are computed at every time-instant



✳ Total error

$$E = \sum_t E(\underline{y(t)}, \hat{y}(t))$$

Cross entropy based error.

Error back propagation

- ✓ Output activations

$$\frac{\partial E}{\partial a^2(t)} = \hat{y}(t) - y(t) \quad \text{for } t = 1 \dots T \quad (I)$$

- ✓ Hidden activations at last instant T

$$\frac{\partial E}{\partial h^1(T)} = (W^2)^T \frac{\partial E}{\partial a^2(T)} \quad (II)$$

- ✓ Hidden activations for previous instances $t = T-1, \dots, 1$ (back propagation)

$$\frac{\partial E}{\partial h^1(t)} = \underbrace{(W^2)^T \frac{\partial E}{\partial a^2(t)}}_t + \underbrace{\frac{\partial h^1(t+1)}{\partial h^1(t)} \frac{\partial E}{\partial h^1(t+1)}}_{(t+1)}$$

Forward Pass

$$\begin{aligned} a^1(t) &= W^1 x(t) + U^1 h^1(t-1) \\ h^1(t) &= \tanh(a^1(t)) \\ a^2(t) &= W^2 h^1(t) \\ \hat{y}(t) &= S(a^2(t)) \end{aligned}$$

$x(1) \dots x(T)$



Error back propagation

- ✓ Hidden activations for previous instances $t = T-1, \dots, 1$

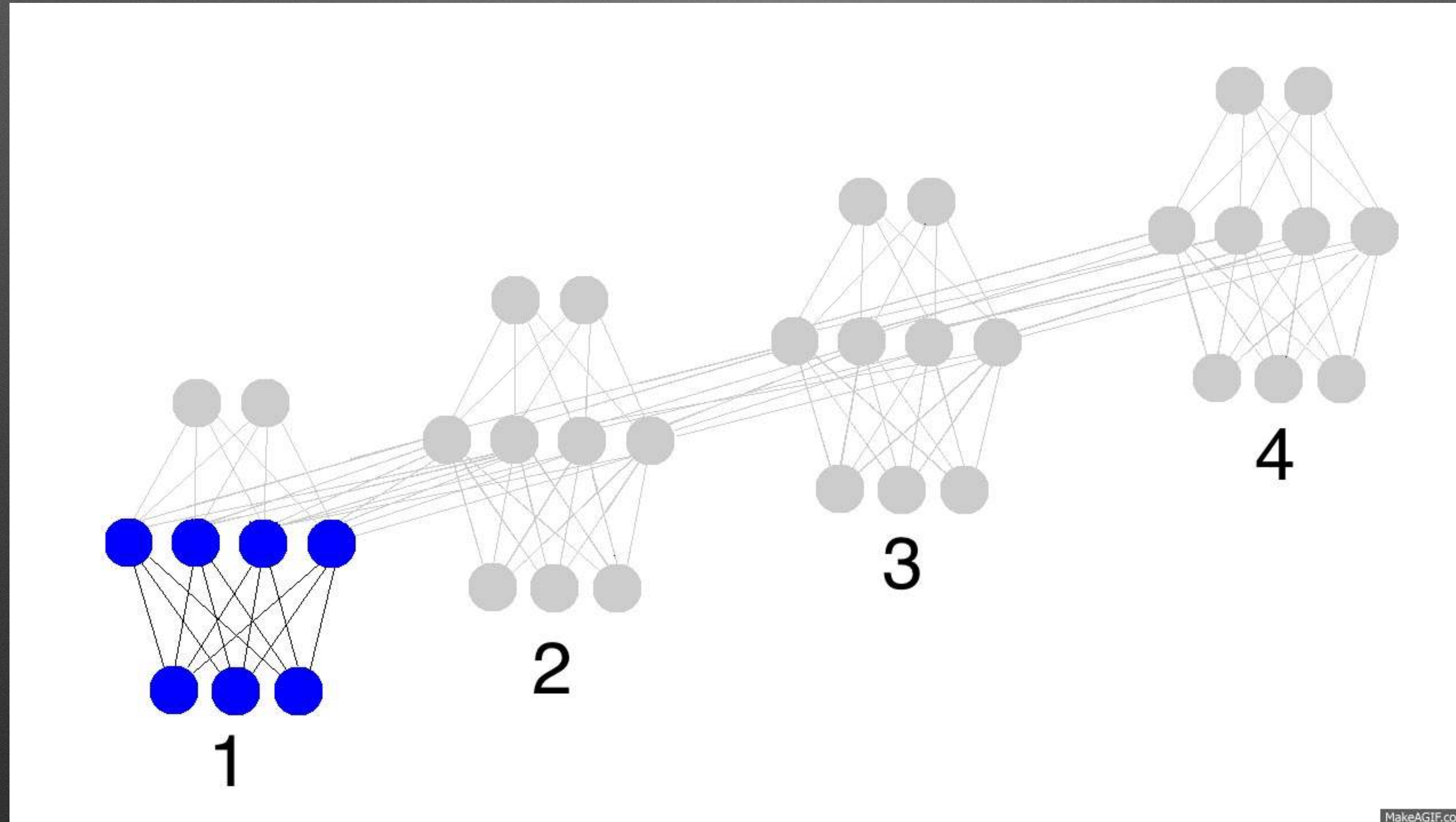
$$\frac{\partial E}{\partial \mathbf{h}^1(t)} = (\mathbf{W}^2)^T \frac{\partial E}{\partial \mathbf{a}^2(t)} + \frac{\partial \mathbf{h}^1(t+1)}{\partial \mathbf{h}^1(t)} \frac{\partial E}{\partial \mathbf{h}^1(t+1)}$$

$$\frac{\partial E}{\partial \mathbf{h}^1(t)} = (\mathbf{W}^2)^T \frac{\partial E}{\partial \mathbf{a}^2(t)} + (\mathbf{U}^1)^T \text{diag}(1 - (\mathbf{h}^1(t+1))^2) \frac{\partial E}{\partial \mathbf{h}^1(t+1)}$$

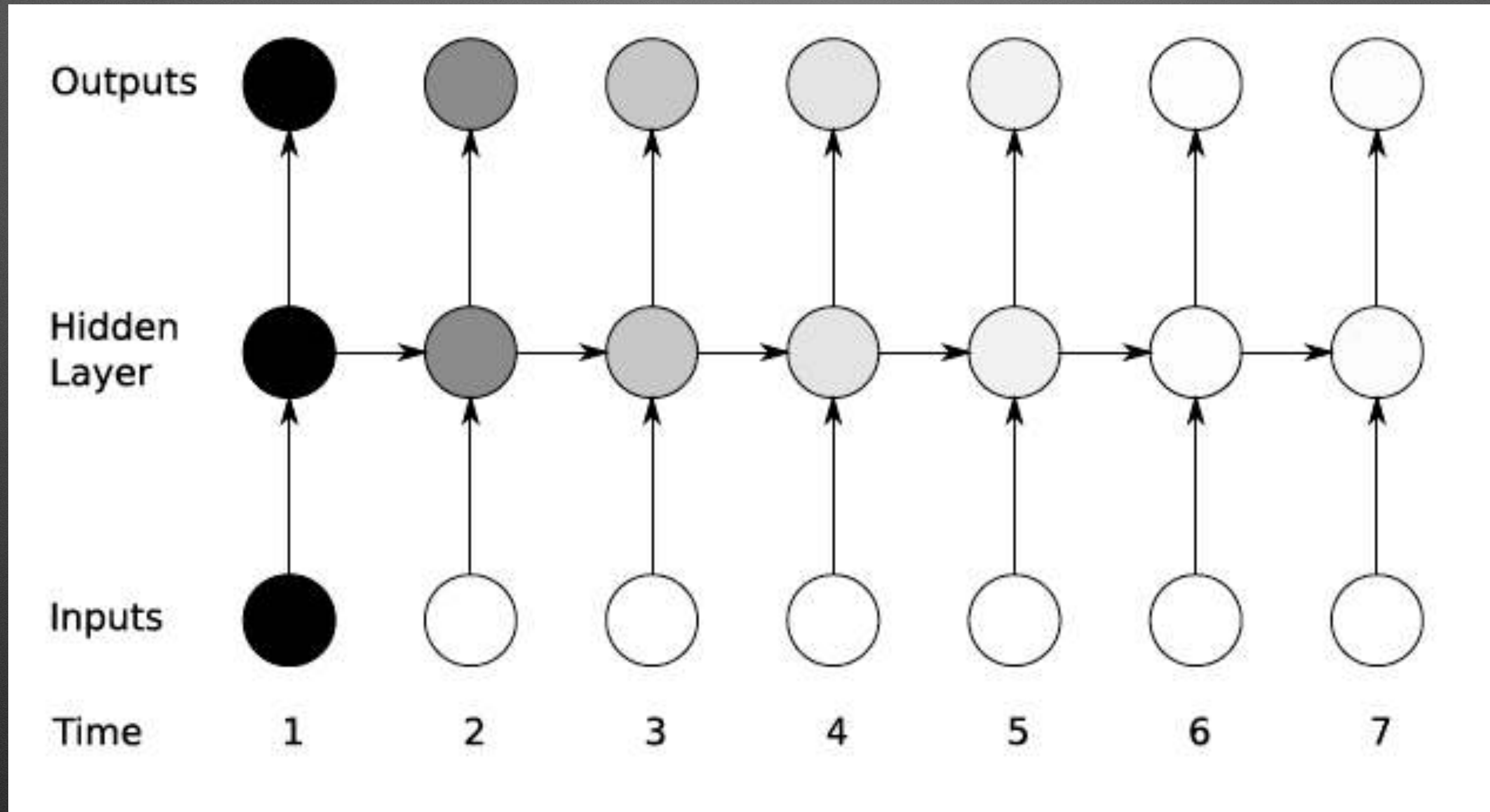
- ✓ Using the above gradient the rest of the gradients can be computed ...



Back propagation through time

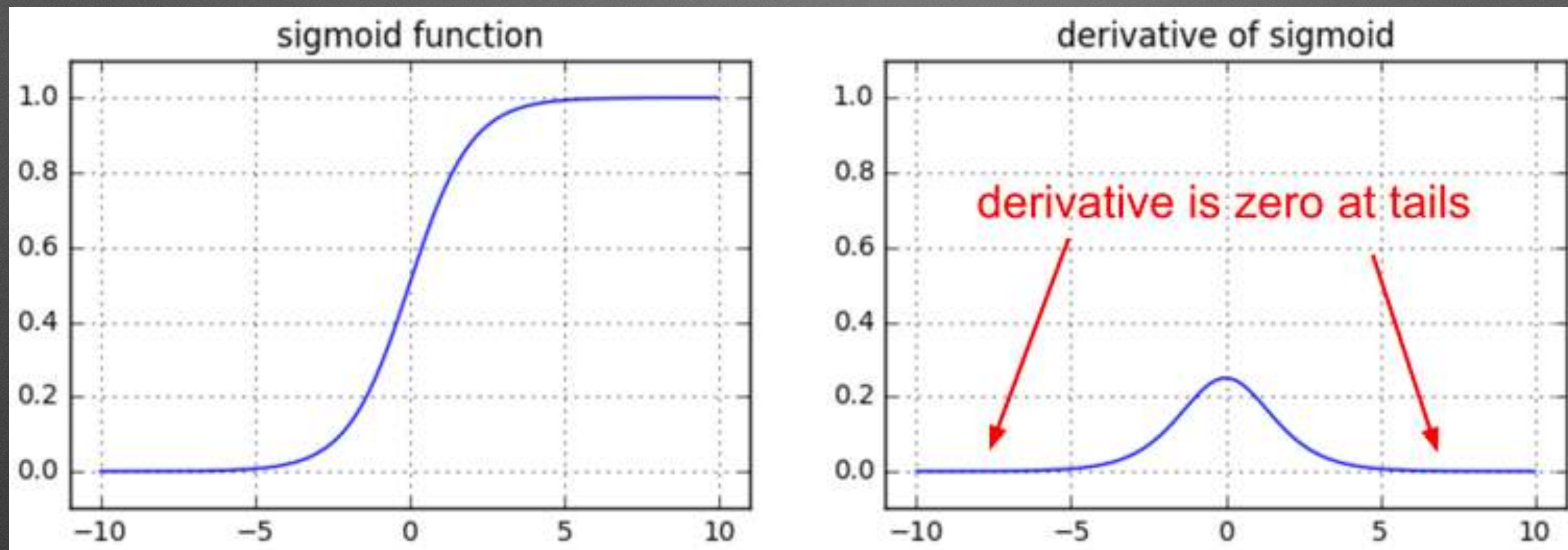


Long-term dependency issues



Long-term dependency issues

- ✓ Gradients tend to vanish or explode



- ✓ Initial frames may not have impact in the final predictions.

Long short term memory (LSTM) idea

